

THANH HUONG TRAN (KYLIE)

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Education

AUT School of Engineering, Computer & Mathematical Sciences, Auckland, NZ May 2026 – Present
Computer Science PhD Student

VNU University of Engineering and Technology, Hanoi, Vietnam Sept. 2022 – Jan. 2025
Master of Computer Science

VNU University of Engineering and Technology, Hanoi, Vietnam Sept. 2018 – June 2022
Bachelor of Electronics and Telecommunications (Advanced Program) *High Distinction*

Research Interests

My research interests are neurological image analysis, multimodal learning and explainable AI. With an emphasis on clinical translation, my study aims to develop interpretable AI technologies that can be applied in real-world healthcare situations.

Publications

Journals

- **Thanh-Huong Tran**, Phuong Anh Nguyen, Anh Ngoc Le, Minh Trien Pham. Optimal CNN Model for Obstructive Sleep Apnea Detection using Particle Swarm Optimization Authors. *Engineering, Technology & Applied Science Research* 15.1 (2025): 19553-19560. **(Q2)**
- Quang Hung Ha, Tri Dung Do, **Thanh-Huong Tran**, Trong-Minh Hoang, Truc Mai Tran, Minh Trien Pham. Life Cycle Stage Prediction for Fall Armyworm: A Case Study from Vinh Phuc Province, Vietnam. *Applied Engineering in Agriculture* 41.6 (2025): 673-679. **(Q3)**
- Hoang Vu Nguyen, Ngoc Bich Dao, **Thanh Huong Tran**, Minh Trien Pham. Using deep learning model to predict micronutrients in food after processing *Vietnam Journal of Science and Technology* version B 66.6 (2024) **(Domestic)**
- **Thanh-Huong Tran**, Hongwei Liang, Wei Li, Minh Trien Pham. Comparison of Machine Learning Algorithms for Induction Motor Rotor Single Fault Diagnosis using Stator Current Signal. *International Journal of Advanced Trends in Computer Science and Engineering* 10.3 (2021): 1509-1514. **(Q4)**

Conferences

- **Thanh-Huong Tran**, Ha Le Thanh, Trung-Nghia Phung, and Minh Trien Pham. OSA Detection using DWT-CNN with Snoring Sounds. In 2024 3rd International Conference on Advances in Information and Communication Technology (ICTA2024), November 16th, 2024, Phu Tho, Vietnam.
- **Thanh-Huong Tran**, Nguyen Xuan Thang, Minh-Trien Pham. A Low-cost Sleeping Mask of Early Screening OSA Support System based on Edge Computing in Smart Home. In 2023 7th International Conference on Engineering Mechanics and Automation (ICEMA 2023), November 11th, 2023, Hanoi, Vietnam.

Granted Patents

- Minh Trien Pham, **Tran Thanh Huong**. A smart system for detect snoring conditions and sleeping positions. National Office of Intellectual Property of Vietnam, 2025.

Research Experience

Robotic Intelligent System Laboratory Sept. 2019 – Feb. 2026
Research Assistant, Teaching Assistant

- Coordinated with team members to establish scientific paper.
- Conducted data analysing via extracting features and classifying the calculation data of not also the electrical devices but other creatures (human, plants, animals, etc.)
- Experienced scientific report writing skills, professional seminars, and international conferences.

Academic Achievements

Prizes & Honours

- AmCham Scholarship Winner by the American Chamber of Commerce in Vietnam (2021-2022)
- Golden Star Award for outstanding academic and work volunteer by Hanoi Student Associations. (2021)
- Distinction Student Award by University of Technology and Engineering, Vietnam National University Hanoi. (2019-2022)

Scholarships

- PhD Scholarships from School of Engineering, Computers and Mathematical Sciences, AUT, Auckland, New Zealand. (For 2026 intake)
- Merit Scholarships from University of Engineering and Technology, Vietnam National University Hanoi. (2019-2022)
- Distinction Student Award, Class of 2018, University of Engineering and Technology, Vietnam National University Hanoi. (2022)

Skills

Foundations: Deep Learning, Computer Vision, Optimization Algorithms

Programming Languages: Python, C/C++

Deep Learning Frameworks: PyTorch, TensorFlow

Computer Vision Frameworks: OpenCV, Scikit-Learn, Scikit-Image

Softwares: Git, Matlab

Languages: Vietnamese (native), Chinese (basic), English (IELTS 7.0)

Academic Services

Reviewer

- ETASR 2026, ICTA 2025

References

- **Assoc. Prof. Catherine Shi** - ✉ - Department of Data Sciences and AI, Auckland University of Technology.
- **Assoc. Prof. Pham Minh Trien** - ✉ - Posts and Telecommunications Institute of Technology.
- **Assoc. Prof. Le Thanh Ha** - ✉ - University of Engineering and Technology, Vietnam National University Hanoi.
- **Dr. Weihua Li** - ✉ - Department of Computer Science, Auckland University of Technology.